

THE CURIOSITY CUP 2025

A Global SAS® Student Competition

Development of Breast Cancer Detection Tool

Machine Learning Dynamite

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ABSTRACT

Purpose

Invasive Ductal Carcinoma (IDC) is the most prevalent form of breast cancer. We utilised a histopathology image database to build a prediction model for identifying IDC images on SAS® Viya® Workbench. The aim of the project is to reduce the discrepancies of cancer diagnoses between pathologists to reduce turnaround times for patients.

Methods

We wrote a code to randomly select 2000 images, 1000 IDC positive and negative, from the database and ran through a series of feature extraction codes via Jupyter extension on SAS® Viya® Workbench. This formed a dataset of 2000 observations, each containing 15 types of extracted features. We input 70% of the dataset (n=1400) into our model development code on SAS® as training data and analysed them through a decision tree model and a HP neural model respectively. We then used the remaining 30% (n=600) for model validation.

Results

Both decision tree and HP neural models performed well in identifying IDC negative and positive images. The decision tree model had an accuracy of 85%, and a false positive rate of 16.7%, false negative rate of 17.5%. While the HP neural model had an accuracy of 89%, 16.1% for false positive rate, and 18.2% for false negative rate.

Conclusions

RGB based image features are promising parameters for developing prediction model for IDC histopathology images and can form the basis for building a model of higher accuracy.

INTRODUCTION

Breast cancer is the most prevalent cancer type in women globally, with Invasive Ductal Carcinoma (IDC) accounting for 83% of breast cancer diagnoses in women and 98% in men (BCRF, 2024). IDC are cancerous cells that have penetrated through the breast ducts and into the surrounding breast tissue (Cancer Research UK, 2023). The diagnosis of IDC relies on histomorphology, requiring breast tissue biopsy and haematoxylin and eosin (H&E) staining that highlights cancerous features to aid pathologist's diagnosis. Haematoxylin stains purple the basophilic nuclear material such as the nucleus and the RNA of the ribosomes, while eosin stains pink the acidophilic cellular structures such as the cytoplasm and cytoplasmic filament. The staining technique allows visualisation of abnormal nuclear structures and cell shape (Ahmed and Abedalthagafi, 2016). However, the process is highly manual and subjective, where the initial and secondary pathologist may have up to 2.3% major and 9.0% minor diagnostic disagreements (Manion, Cohen and Weydert, 2008). The highly subjective process leads to delay of diagnosis and treatment for the patient, which is undesirable considering the high progression rate of IDC.

We chose the 'Breast Histopathology Images' dataset, published by Paul Mooney at Kaggle, a machine learning and data science community (Mooney, 2017). The image data originate from 162 IDC mount slide images, captured at 40x magnification and extracted per 50x50 pixel to form 277,524 patches, with 198,738 negatives for IDC and 78,786 positives. Janowczyk and

Madabhushi (2016) utilised the database to create a deep learning model, where we aim to also use the set to develop a classification model via different parameters and models (Janowczyk and Madabhushi, 2016). Based on Kunhoth and Al-Maadeed's study (2024), image features such as RGB values are promising tools for classifying cancer histopathology images (Kunhoth and Al-Maadeed, 2024). We aim to utilise these colour information and other image details to identify correlations and apply deep learning to understand how these features help predict IDC class.

The development of an IDC prediction model allows objective insights to diagnosis, as well as the ability to process large quantities of histopathology samples. The clinical application of the model also serves as a second opinion for pathologists, allowing more efficient and accurate diagnosis.

Feature Extraction from Images

We analysed the dataset uses a Python extension to extract key image features from histopathological samples to aid in tissue classification and cancer detection. The extracted features include RGB colour components, Gary-Level Co-occurrence Matrix (GLCM) texture features, and histogram-based statistical measures. These features were selected based on their relevance in identifying morphological and chromatic differences in stained tissue samples. The target variable in this dataset is the classification of the images into either cancerous (1) or non-cancerous (0), which forms the basis of our predictive modelling task.

We used the RGB colour model to measure the red, green, and blue intensities of each image. Studies have shown that variations in RGB colour intensity can indicate malignancies due to increased nuclear staining (Wahyuni, Tiyas, and Murnani, 2022). The HSV colour model was also used to analyse image features in a way that aligns more closely with human perception. Hue represents the dominant wavelength of colour, saturation measures the intensity of colour, and value reflects brightness (Wahyuni, Tiyas, and Murnani, 2022).

GLCM texture analysis was applied to capture spatial relationships between pixel intensities. This method computes four key texture properties: contrast, correlation, energy, and homogeneity. Contrast measures the intensity differences across neighbouring pixels, correlation assesses texture uniformity, energy represents texture smoothness, and homogeneity indicates local similarity. These features are crucial for identifying structural differences between normal and cancerous tissues. Research suggests that GLCM-based texture features effectively distinguish between tissue types in medical imaging (Kothari et al., 2013).

Histogram-based statistical features, including mean, variance, standard deviation, and skewness, were also extracted. These metrics describe the distribution of pixel intensities within an image, providing insights into brightness, contrast, and asymmetry. Such features help differentiate between dense and loosely packed cellular structures in histopathology. Previous studies indicate that variations in histogram features correspond to changes in tissue organisation and can be used for automated pathology detection (Lu et al., 2019)

We were able to extract 15 features from our image analysis, which served as the foundation for predictive modelling, enabling automated analysis and classification of cancerous and non-cancerous tissue samples.

Model development

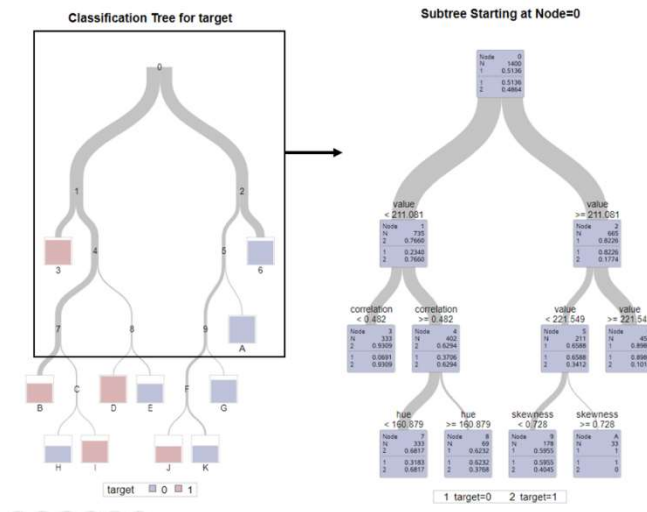
Decision tree model

In machine learning, a decision tree is a mathematics-based classification method to sort datasets with relevant variables (Lu, Ye, and Zheng, 2022). We constructed the decision tree model using SAS® language, where `proc hpsplit` created classes for the dataset and further

analysed via the Gini split criterion, a metric for calculating classification purity (Lu, Ye, and Zheng, 2022). The development codes are as below:

```
proc hpsplit data=mylib.train;
  class target;
  model target(event='1') = mean_r mean_g mean_b color_ratio contrast
                           correlation energy homogeneity hue saturation
                           value mean variance std_dev skewness;
  grow gini;
  prune costcomplexity;
  output out=mylib.dtree_train_pred;
  code file='/workspaces/workspace/.sasuser.workbench/dtree_score.sas';
run;
```

The model reviewed and prioritised variables according to their Pearson Correlation Coefficients, correlating to the IDC classes. Data entries were filtered based on mathematical thresholds to form branches of the decision tree, which was continued further down the decision tree to form a tree depth of 5 layers and 11 leaves after data pruning (Figure 1). The model found the variable 'value' had the most importance to the decision tree, followed by 'correlation' and 'skewness'.



3 layers of the tree was presented from its origin node 0, indicating how differentiation of datapoints were conducted based on highly related variables and mathematical thresholds; 'target 0' / '1' indicates Invasive Ductal Carcinoma (IDC) negative entries, while 'target 1' / '2' indicates IDC positive entries.

Testing dataset (n=600) was then inputted to validate the model. Receiver Operating Characteristic (ROC) curve visualised the model performance with Area Under Curve (AUC) value, indicating how well the model differentiate IDC positive and negative inputs (Figure 2).

Figure 1. Structure of decision tree diagram.

Training data (n=1400) achieved an AUC of 0.89, and 0.85 for testing data (n=600).

Overall, our model achieved highly significant results. Model fit statistics indicate a better fit outcome than the intercept-only-model,

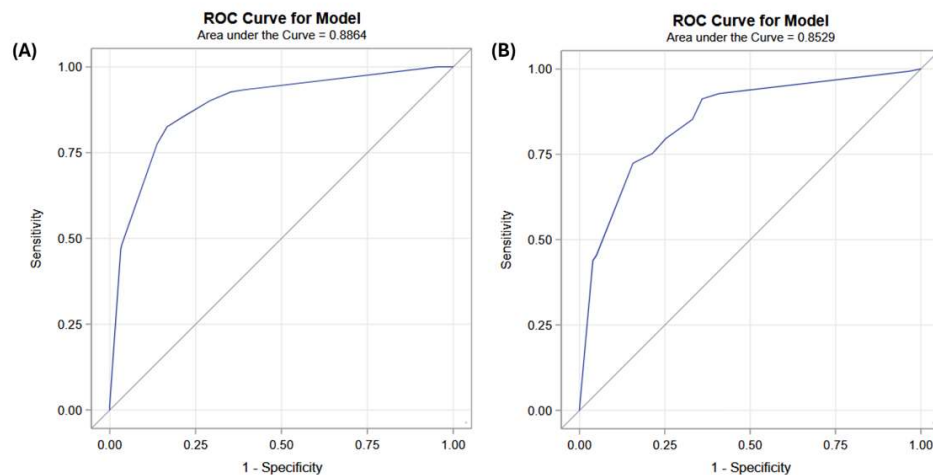


Figure 2. ROC curve for model.

with a lower Akaike Information Criterion of 581.71 than 831.37 of the compared models. Results of the likelihood ratio test, $\chi^2 = 251.6556$, $p < .0001$, also suggested that the model provided a better fit than a null model. In addition to the 85% model accuracy, false positive rate of 16.7%, and false negative rate of 17.5%, this indicate that the model is effective in distinguishing between IDC classes, achieving high reliability.

HP neural model

A HP neural reads data as a multilayer network, where it passes predictions through layers of information to map classification or predictions (SAS® Enterprise Miner™, 2022). We developed the HP neural network with the `proc hpneural` procedure, which learns independently related features of the dataset through multiple hidden layers (MLP architecture). Since HP neural conducts independent learning, the procedure further divides the 1400 inputs into 1050 training data and 350 validation data for deep learning. The development codes are as below:

```
proc hpneural data=mylib.train_final;
    target target / level=nominal;
    input mean_r mean_g mean_b color_ratio contrast
        correlation energy homogeneity hue saturation
        value mean variance std_dev skewness / level=interval; hidden 10;
    train outmodel=mylib.hpneural_model;
    code file='/workspaces/workspace/.sasuser.workbench/hpneural_score.sas';
    score out=mylib.train_scored;
run;
```

The multilayer learning approach allows the model to learn independently the best combination for prediction, producing the best classification outcome. An iteration table illustrates how this process is conducted (Figure 3). Best iteration achieved at 50th iteration, with average training error of 0.360053 and average validation error of 0.391835. Testing data (n=600) were inputted, where the model has an AUC of 0.89 and 10.9% discordant (Figure 4). Overall, the model showed high accuracy for classification and prediction.

Iteration Table			
Iteration	Avg Training Error	Avg Validation Error	Best?
0	0.695496	0.695705	
1	0.687328	0.686788	
2	0.607008	0.608286	
3	0.597799	0.603922	
4	0.542617	0.547237	
5	0.536446	0.536302	
6	0.532996	0.534350	
7	0.484735	0.496696	
8	0.448087	0.480315	
9	0.444103	0.480570	
10	0.442356	0.477796	

Figure 3. Iteration table for finding the best combination.

Model accuracy

Both models showed ability to differentiate IDC negative and positive images based on extracted features. The HP neural model performed better than the decision tree model in terms of percentage concordant, achieving more accurate prediction despite having less learning data (Table 1).

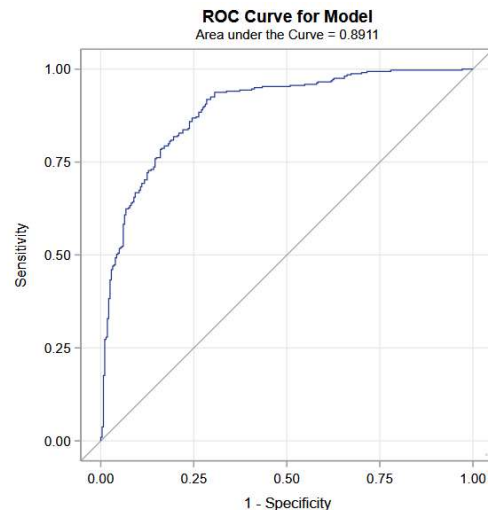


Figure 4. ROC curve for testing data.

Model validation	False positive rate	False negative rate	Overall Misclassification rate
Decision tree (n=600)	16.7%	17.5%	15%
HP neural (n=350)	16.1%	18.2%	11%

Table 1. Comparison of misclassification rate based on validation data

CONCLUSION

Our model has showed its potential to become a promising clinical tool, which is useful to facilitate pathologist's diagnosis of IDC. Further improvements of the model may include adding more parameters into the predictive model, allowing more layers of learning via the multilayer neural network to achieve enhanced accuracy.

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